

Human body showing symmetry



Sponge showing asymmetry

2.2 SYMMETRY

The concept of symmetry is fundamental in understanding the organisation of an animal. Symmetry in animals is balanced distribution of paired body parts. The body plan of a vast majority of metazoans exhibits some kind of symmetry. However, most of the sponges and snails show asymmetry (lack of symmetry). The symmetry of an animal and its mode of life are *correlated*.

2.2.1 Asymmetry

The animals, which cannot be cut into two equal parts (antimeres) in any plane passing through the centre of the body are called asymmetrical, e.g. most sponges and adult gastropods. In the asymmetrical animals, the body lacks a definite form. Asymmetry cannot be said to be an adaptation or advantage to an organism. Primarily asymmetrical organisms do not develop complex sensory and locomotor functions.

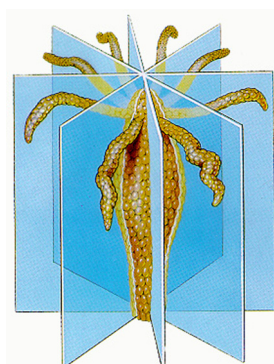
2.2.2 Symmetry

The regular arrangement of body parts in a geometrical design relative to the axis of the body is called symmetry. The animals, which can be cut into two equal parts, or antimeres in one or more planes passing through the '**principal axis**' of the body are called symmetrical animals. In a symmetrical animal, paired body parts are arranged on either side of the plane passing through the principal axis, such that they are equidistant from the plane. The unpaired body parts are located mostly on the plane, passing through the principal axis. Basically, the symmetry in animals is of two kinds:

- i) **Radial symmetry** and
- ii) **Bilateral symmetry**

i) **Radial Symmetry or Monaxial heteropolar (axis is single; poles are different) Symmetry**

When any plane passing through the central axis (**oro-aboral axis/ principal axis**) of the body divides an organism into two identical parts, it is called **radial symmetry**. The animals with radial symmetry are either sessile or planktonic or sluggish forms. It is the principal symmetry of the diploblastic animals such as the cnidarians and ctenophores (considered as biradial animals by some authors). Animals showing radial symmetry live in water and they can respond equally to stimuli that arrive from all directions. Thus, radial symmetry is an advantage to sessile or slow moving animals. However, triploblastic animals such as echinoderms are 'secondarily radially



Hydra- showing radial symmetry

symmetrical' (as it is five angled, it is also called **pentamerous radial symmetry**). Radially symmetrical animals have **many planes** of symmetry, whereas pentamerous radially symmetrical animals have **five planes** of symmetry.

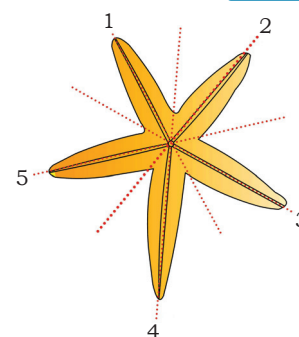
ii) Bilateral symmetry

When **only one plane** (*median sagittal plane*) that passes through the central axis (anterior – posterior axis) divides an organism into two identical parts, it is called **bilateral symmetry**. It is the '*principal type of symmetry*' in the **triploblastic** animals. Among the triploblastic animals, some **gastropods** become secondarily **asymmetrical**, though they have primarily bilaterally symmetrical larvae.

Symmetry of an organism is determined based on the symmetry of the early developmental stages and not that of the adults.

Bilaterally symmetrical animals are more efficient than the other animals in seeking food, locating mates and in avoiding or escaping from predators, because of **cephalization** (*concentration of nerve and sensory cells at the anterior end*). As a result of cephalization, bilaterally symmetrical animals can sense the new environment into which they enter and respond more efficiently and quickly.

Note: Sea anemones and acnidarians are said to show biradial symmetry.



Sea Star showing pentamerous symmetry

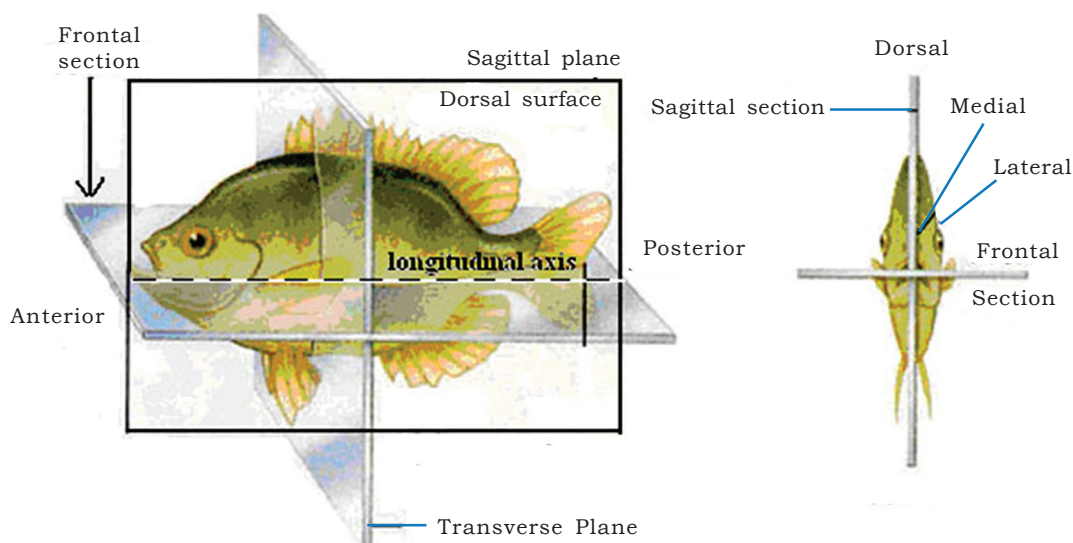


Fig. 2.4 Fish showing bilateral symmetry